TERTIARY INFOTECH ACADEMY

LAB 11
Diagnose Storage and Database Performance, Then Recover

AWS Certified SysOps Administrator Associate Training
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Scenario
A transactional workload shows rising latency after a change, and the operator must determine whether compute, storage, connections or query behaviour is the bottleneck.

Objective
Apply K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.

Deliverable
A metric analysis, bottleneck decision, change plan, restore checklist and database recovery evidence.

Services and tools
CloudWatch · EBS · RDS · Performance Insights · AWS Backup

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Open `data/metrics.csv`; define units, timestamps and expected baseline before calculating any conclusion.
```
column -s, -t data/metrics.csv
```

Evidence note: _____

Step 2 — Run `analyse_metrics.py`; inspect the output ranking and confirm the highest deviation is reproducible.
```
python3 analyse_metrics.py | tee evidence/metric-analysis.txt
```

Evidence note: _____

Step 3 — Compare CPU, EBS queue, IOPS, throughput, connections and latency in `BOTTLENECK-DECISION.md`.
```
open BOTTLENECK-DECISION.md
```

Evidence note: _____

Step 4 — Inspect the learner RDS configuration and CloudWatch metrics for the same interval where available.
```
aws rds describe-db-instances --db-instance-identifier REPLACE_DB --output json | tee
evidence/rds-config.json
```

Evidence note: _____

Step 5 — Review Performance Insights or database-engine evidence for waits, slow statements or locks; record limitations if unavailable.

```
open QUERY-EVIDENCE.md
```

Evidence note: _____

Step 6 — Propose one reversible action with risk, owner, expected signal and rollback; do not change multiple variables.

```
open CHANGE-PLAN.md
```

Evidence note: _____

Step 7 — Create a snapshot or approved recovery point and record its timestamp and status.

```
aws rds create-db-snapshot --db-instance-identifier REPLACE_DB --db-snapshot-identifier
sysops-lab11-REPLACE_SUFFIX | tee evidence/snapshot.json
```

Evidence note: _____

Step 8 — Restore to an isolated learner identifier and verify engine status and a trainer-provided data check.

```
aws rds restore-db-instance-from-db-snapshot --db-instance-identifier sysops-lab11-restore-
REPLACE_SUFFIX --db-snapshot-identifier sysops-lab11-REPLACE_SUFFIX | tee
evidence/restore.json
```

Evidence note: _____

Step 9 — Capture metrics, snapshot and restore evidence as JPEGs; record actual recovery timing.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence note: _____

Step 10 — Run validation; delete the isolated restore and learner snapshot when approved.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.

Acceptance criteria

- Units and baseline stated
- Cause remains a hypothesis until tested
- One variable changed
- Restore does not overwrite source
- Data validation and actual timing recorded

Evidence checklist

□ Metric analysis

□ RDS configuration

□ query evidence

□ snapshot/restore records

□ completed bottleneck and change decisions

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.